

Field Lab

Immigration, Foreign Investment, and Infrastructure Shocks in Lisbon's Housing Market: A Causal Machine Learning and Policy Evaluation Approach

A – Background

Minimum number of students: 3

Maximum number of students: 4

This project evaluates the impact of immigration inflows, foreign investment, and infrastructure expansion on housing prices and transaction volumes in Lisbon.

The study combines administrative and market data from INE, Banco de Portugal, land registry transactions, migration statistics, and web-scraped real estate listings to build a causal machine learning framework capable of identifying nonlinear and spatial effects.

The goal is to determine whether foreign capital inflows, workforce immigration, and infrastructure investments amplified housing price growth, inequality, and speculative pressures in Lisbon.

Keywords: Business Analytics; Machine Learning

B - Objectives of the project

The main goal is to estimate the causal and heterogeneous effects of immigration, foreign investment, and infrastructure shocks on Lisbon's housing market using advanced causal machine learning techniques.

The project will:

- Quantify the impact of immigration inflows on housing prices and rents
- Evaluate the causal effects of the Golden Visa reform
- Measure the effect of infrastructure expansion on local property values
- Identify nonlinear interactions between immigration and infrastructure
- Estimate heterogeneous treatment effects across neighborhoods
- Conduct counterfactual simulations (e.g., no Golden Visa reform, delayed infrastructure projects)

- Assess implications for housing affordability and inequality

C- Methodology and recommended/required tools

Data Collection Method

- Regional housing prices and demographic indicators from INE
- Mortgage flows and credit conditions from Banco de Portugal
- Land registry transaction-level data
- Migration statistics (AIMA/administrative sources)
- Web-scraped listings (prices, rents, time-on-market)

Data Analysis

- Double Machine Learning (DML) for causal effect estimation
- Causal Forests to estimate heterogeneous treatment effects
- Spatial Machine Learning models to capture spillover effects
- Agent-Based Modeling (ABM) simulating foreign vs. domestic buyer competition

Recommended Tools

- Python

D- Format and End product / Deliverables:

- Final academic thesis report
- Replicable ML pipeline documentation
- Causal impact visualization dashboard (optional advanced output)

Division Between Collective vs. Individual Work

Collective Work:

- Research design and identification strategy
- Data collection and cleaning
- Model integration and validation
- Policy interpretation and final presentation

Individual Work:

- One student: Causal ML implementation (DML / Causal Forests)
- One student: Spatial ML and infrastructure impact modeling

- One student: Agent-Based simulation design
- One student (if applicable): Data engineering and web scraping pipeline

E - Work Plan

Dates - Start: September 2026 **End:** December 2026

Number of students: 3-4

Meetings: weekly (online)

F - Readings and support materials

- **Causal Machine Learning (Double ML, Causal Forests)**
- **Difference-in-Differences and Event Study methodologies**
- **Urban economics and housing market dynamics**
- **Immigration and housing demand literature**
- **Spatial econometrics and geospatial ML**
- **Policy evaluation frameworks**

G – Roles and responsibilities

Students will:

- Conduct literature review
- Design identification strategy
- Implement machine learning models
- Perform simulations and robustness checks
- Draft thesis and policy report
- Present findings to faculty panel

The advisor will:

- Provide methodological guidance in ML and causal inference
- Review coding and model implementation
- Ensure econometric rigor
- Provide feedback on drafts
- Support institutional data access (if applicable)

H – Advisors

Advisor / Coordinator: Miguel Matos

I – Other information

Advisor short bio notes

Miguel Matos | Professor of Computer Programming at Nova SBE & Software Engineer. Specialized in Machine Learning, Computer Vision, and Web Development.

Application

Please send an e-mail to miguel.matos@novasbe.pt expressing interest, including:

- Academic transcript
- Brief motivation statement

Outcome of application

Selected students will be notified by email and invited to an initial project planning meeting.